L_0-Based Sparse Modeling

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Contents

0.1 l_0 quasi-norm

$\|x\|_0$: the number of nonzero components of the vector $x \in \mathbb{R}^n$.

0.2 l_0, l_2 combined

$$\|x\|_0 + \frac{1}{\delta^2} \|x\|_2^2 \tag{1}$$

$\frac{1}{\delta^2} \|x\|_2^2$: an elastic potential energy?Is there an interpretation?(Bernoulli-gaussien from Soussen and references therein)

0.3 l_0 vs l_0, l_2 combined

$\|.\|_0 + \frac{1}{\delta^2} \|.\|_2^2$ is an upper approximation of $\|.\|_0$ and it converges toward it in some "sense"(Soussen, Bertsimas et al.) when $\delta \to +\infty$.

0.4 Questions

- Is there cases where sparse modeling with $\|.\|_0 + \frac{1}{\delta^2} \|.\|_2^2$ is "better" or "more appropriate" than $\|.\|_0$ solely? In what sense? Where? When? How?
- Is sparse modeling with $\|.\|_0 + \frac{1}{\delta^2} \|.\|_2^2$ more "tracktable" than $\|.\|_0$ from the point of view of mathematical programming?

References

Bibliography

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